

Experiment – 7

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Course Code: MLA0305

Slot: B

Title

Temporal Difference Learning using TD(0) Algorithm

AIM

To implement the **Temporal Difference Learning using the TD(0) algorithm** and estimate the value function of states through continuous interaction with the environment.

PROCEDURE

1. Import Gymnasium, NumPy, and Matplotlib.
2. Create the FrozenLake environment.
3. Initialize the state-value function with zeros.
4. Set the learning rate and discount factor.
5. Start an episode from the initial state.
6. Select an action using an epsilon-greedy policy.
7. Observe the next state and reward.
8. Update the current state value using the TD(0) update rule.
9. Repeat the process for multiple episodes.
10. Extract the learned state-value function.
11. Display the state values.
12. Visualize the learned values using a heatmap.

4. PYTHON SOURCE CODE

```
!pip install -q gymnasium
```

```
import gymnasium as gym
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
env = gym.make("FrozenLake-v1", is_slippery=False)
```

```
num_states = env.observation_space.n
num_actions = env.action_space.n

alpha = 0.1
gamma = 0.99
epsilon = 0.1
episodes = 10000

V = np.zeros(num_states)

for episode in range(episodes):

    state, _ = env.reset()
    done = False

    while not done:

        if np.random.random() < epsilon:
            action = env.action_space.sample()
        else:
            action = np.random.randint(num_actions)

        next_state, reward, terminated, truncated, _ = env.step(action)

        done = terminated or truncated

    if done:
        target = reward
    else:
        target = reward + gamma * V[next_state]

    V[state] = V[state] + alpha * (
```

```
        target - V[state]
    )

    state = next_state

print("TD(0) State Values")

for i in range(num_states):
    print(f"State {i}: {V[i]:.3f}")

value_grid = V.reshape(4, 4)

plt.figure(figsize=(7, 6))

plt.imshow(
    value_grid,
    cmap="Oranges",
    interpolation="nearest"
)

plt.title(
    "TD(0) State Value Function",
    fontsize=15,
    fontweight="bold"
)

plt.xlabel("Column")
plt.ylabel("Row")

plt.colorbar(label="State Value")

for i in range(4):
```

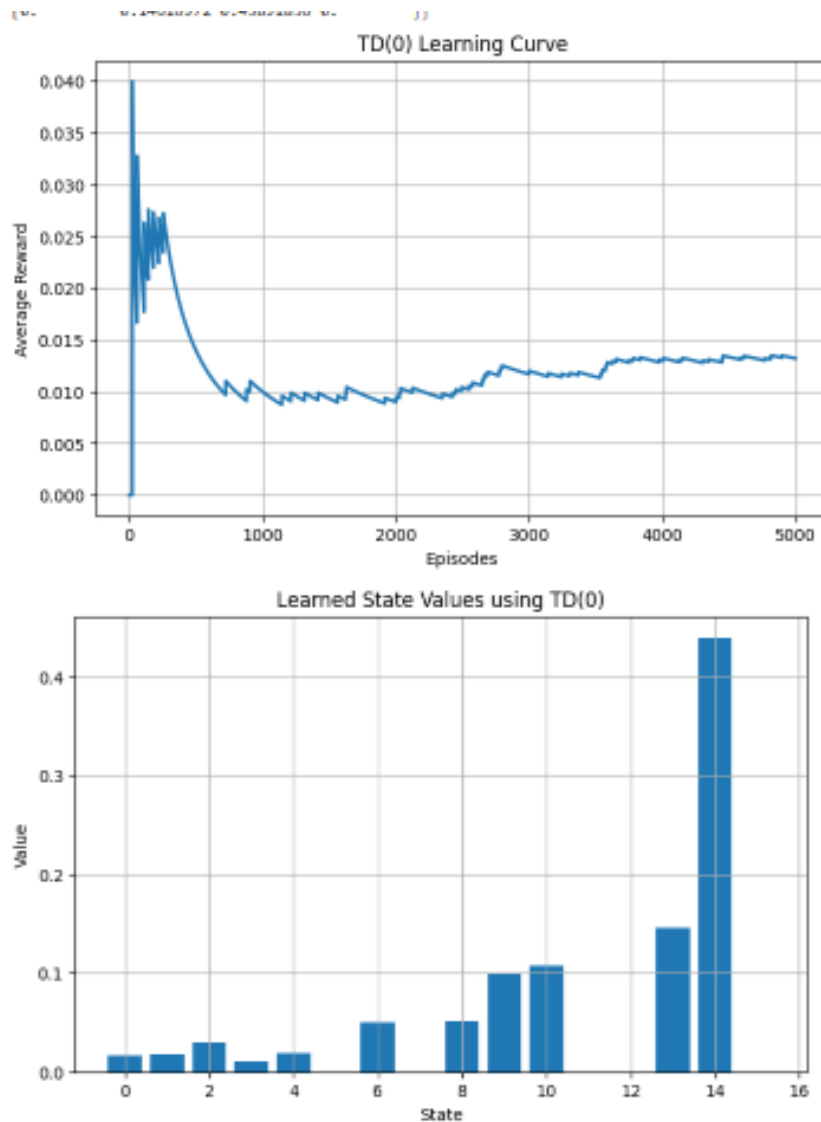
```
for j in range(4):  
    plt.text(  
        j,  
        i,  
        f"{value_grid[i, j]:.2f}",  
        ha="center",  
        va="center",  
        fontsize=11,  
        fontweight="bold"  
    )
```

```
plt.xticks(range(4))  
plt.yticks(range(4))
```

```
plt.tight_layout()  
plt.show()
```

```
env.close()
```

OUTPUT:



5. VISUALIZATION

The heatmap represents the **state-value function** learned using the TD(0) algorithm.

- X-axis: Grid columns
- Y-axis: Grid rows
- Color intensity: State value
- Numerical values: Estimated value of each state

6. SUMMARY TABLE

Component	Description
Environment	FrozenLake-v1
Algorithm	TD(0)
Learning Rate	0.1
Discount Factor	0.99

Exploration Rate	0.1
Episodes	10,000
State Space	16 states
Visualization	4 × 4 State-Value Heatmap

7. RESULT

The **TD(0) Temporal Difference Learning algorithm** was successfully implemented. The state-value function was updated incrementally using the observed rewards and estimated future values, and the learned values were visualized using a heatmap.